Survey Design Matrix — Blank Template

Fillable PDF. Click a blue box to type; tick the small squares. Save with **File › Save** and attach the saved PDF to your assignment. Works best in the free Adobe Acrobat Reader; macOS Preview and web browsers also work. If an answer does not fit its box, continue it on an extra page. Source: `Step_06_Instrument_Construction/templates/survey_matrix_template.md` (rebuild with `tools/make_fillable_pdfs.py`).

For Socio-Ecological and Environmental Research

Instructions for use.

Fill in every row before building your survey instrument. If you cannot complete the "Research Question" or "Analytic Use" columns for an item, that item should be cut or revised. This matrix is a living document — update it after cognitive interviews, piloting, and peer review.

Save a dated version (`survey_matrix_v1_YYYY-MM-DD.md`) at each major revision.

Part 1 — Study Overview

Field	Your entry
Study title	
Lead researcher	
Date (version)	
Study site / population	
Target population	
Survey mode	☐ Online ☐ Paper ☐ Face-to-face (tablet) ☐ Phone ☐ Mixed
Sampling strategy	
Estimated duration	
Primary research questions	RQ1: RQ2: RQ3:
Pre-registration DOI / link	
IRB / Ethics approval #	

Field	Your entry
Stimulus materials (if any)	

Part 2 — Full Item Matrix

Add or remove rows as needed. Group items by survey block/page.

Block 1 — [Block name, e.g., Introduction / Consent]

Block ID	Item text (verbatim or draft)	Research Question(s) addressed	Theoretical Construct	Variable Role	Scale / Prior Source	Response Format	Placement / Order Notes	Justification / Cut rationale
B1-01				☐ Not in analyses ☐ IV ☐ DV ☐ Covariate ☐ Random effect ☐ Qualitative				
B1-02								

Block ID	Item text (verbatim or draft)	Research Question(s) addressed	Theoretical Construct	Variable Role	Scale / Prior Source	Response Format	Placement / Order Notes	Justification / Cut rationale

Need more rows? Add a continuation page or attach a spreadsheet with the same columns.

Block 2 — [Block name]

Block ID	Item text	Research Question(s)	Theoretical Construct	Variable Role	Scale / Source	Response Format	Placement Notes	Justification
B2-01								

Need more rows? Add a continuation page or attach a spreadsheet with the same columns.

Block N — [Add as many blocks as needed]

Part 3 — Variable Summary Table

Populate after the full matrix is complete. Every variable in your analysis plan must appear here.

Variable name	Role	Type	Block ID(s)	Scale/range	Missing data plan	Notes
	☐ IV ☐ DV ☐ Covariate ☐ Random effect	☐ Binary ☐ Nominal ☐ Ordinal ☐ Continuous				

Need more rows? Add a continuation page or attach a spreadsheet with the same columns.

Part 4 — Respondent Burden Audit

Survey block	Estimated time	Cumulative time
Block 1 —		

Survey block	Estimated time	Cumulative time
Block 2 —		
Block N —		
Total		

Benchmark targets:

- Online / self-administered: ≤ 15 min preferred; ≤ 20 min acceptable
- Face-to-face with enumerator: ≤ 35 min
- Phone: ≤ 10 min
- With multimedia stimuli (photos, audio, video): add ~1–2 min per stimulus block

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Part 5 — Item-level QC Checklist

For every item in the matrix, verify:

- [ ] The item maps to at least one research question
- [ ] The item uses an established scale OR construction rationale is documented
- [ ] The response options are **exhaustive** (cover all realistic answers)
- [ ] The response options are **mutually exclusive**
- [ ] The item is free of **double-barrelled** language (asks one thing at a time)
- [ ] The item avoids **leading** or **loaded** phrasing
- [ ] The item is readable at the education level of the target population
- [ ] Sensitive items (income, race/ethnicity, gender, health) are placed at the **end**
- [ ] Categories for sensitive items include **inclusive options** (e.g., non-binary, prefer not to say, other)
- [ ] Open-ended items have a clear **character limit** and a purpose that justifies the added burden
- [ ] If order is unconstrained, **randomization** within the block is planned and pre-registered

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Part 6 — Question-format Decision Guide

Use the table below to select the appropriate format for each item.

Format	Best for	Burden	Inferential strength	Notes
Likert scale (agree–disagree, 5–7 pt)	Attitudes, beliefs, values	Low	Moderate–high	Use odd-point scale unless mid-point is theoretically meaningful to exclude

Format	Best for	Burden	Inferential strength	Notes
Semantic differential	Bipolar affective dimensions (e.g., like/dislike)	Low	Moderate–high	Anchors must be true opposites
Ranking	Relative preference among a small set (≤ 7 items)	Moderate	Moderate	Respondents cannot express equal preference; avoid for > 7 items
Slider / Visual Analogue Scale	Continuous preference, willingness to pay	Low–moderate	High	Requires digital delivery; avoids artificial ordinality
Multiple-choice (single answer)	Demographics, category membership	Very low	Categorical only	Ensure "other" + "prefer not to say" options
Multiple-select (check all that apply)	Behaviours, multiple identities	Low	Categorical only	Avoid if items are not independent
Word association / free-list	Cultural salience, mental models	Moderate	Qualitative	Analyse with cultural consensus or salience indices
Short text (1–3 sentences)	Mechanism, justification	High	Qualitative	Use only where depth is essential
Open-ended (long text)	Exploration, narrative	Very high	Qualitative	Consider alternatives; causes dropout
Attention check	Data quality / bot detection	Very low	Procedural	Frame as comprehension aid with community partners
Forced-choice / best-worst scaling	Preference without scale use bias	Moderate	High	Useful for cross-cultural comparisons

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Part 7 — Demographic Item Bank

Copy relevant items into your matrix. Adapt categories to your study population.

Age

"What is your age?" _____ years

Or: ☐ 18–24 ☐ 25–34 ☐ 35–44 ☐ 45–54 ☐ 55–64 ☐ 65+ ☐ Prefer not to say

Gender

"What is your gender identity?"

☐ Woman ☐ Man ☐ Non-binary / gender non-conforming ☐ Two-Spirit ☐ Prefer to self-describe: _____ ☐ Prefer not to say

Race / Ethnicity ***(adapt to country / context)***

"Which of the following best describes your racial or ethnic background? (Select all that apply)"

[Add country-appropriate categories] ☐ Prefer not to say

Education

"What is the highest level of education you have completed?"

☐ No formal schooling ☐ Some primary school ☐ Completed primary ☐ Some secondary ☐ Completed secondary ☐ Vocational / technical ☐ Some university ☐ University degree ☐ Graduate degree ☐ Prefer not to say

Household income *(sensitive — place at end; consider necessity)*

"Which range best describes your total annual household income (before taxes)?"

[Insert country-appropriate ranges] ☐ Prefer not to say

Place of residence / tenure

"Do you currently live in [community/region]?" ☐ Yes ☐ No

"How many years have you lived here?" _____ years

Occupation / Livelihood

"Which of the following best describes your primary occupation?"

☐ Agriculture / farming ☐ Fishing / aquaculture ☐ Forestry ☐ Tourism / hospitality ☐ Government / public sector ☐ NGO / non-profit ☐ Business / commerce ☐ Student ☐ Retired ☐ Unemployed / seeking work ☐ Other: _____ ☐ Prefer not to say

Proximity / dependence on natural resources *(socio-ecological studies)*

"How much does your household depend on [forest / river / land] for your income or food?"

☐ Not at all ☐ A little ☐ Somewhat ☐ A lot ☐ Almost entirely

Part 8 — Attention Check Examples

Include ≥ 2 attention checks in surveys > 10 min. Place one in the first third and one in the final third.

Type 1 — Instructed response item

"*This is a question to make sure you are reading carefully. Please select '**Somewhat agree**' for this item.*"

☐ Strongly disagree ☐ Disagree ☐ Neutral ☐ Somewhat agree ☐ Strongly agree

Type 2 — Screener / nonsense item

"*In the past week, how often did you fly to the moon?*"

☐ Never ☐ Once or twice ☐ Several times ☐ Daily *(correct answer: Never)*

Type 3 — Speeder flag *(platform-level, not item-level)*

Flag responses with total completion time < [expected time × 0.4] for review.

Note for community-based research: Frame attention checks as comprehension aids:

"*We include a few questions like this to make sure the survey is working correctly on your device.*"

Do not frame them as "traps" or imply distrust. See Abbey & Meloy (2017).

Part 9 — Placement and Order Principles

Principle	Rule	Rationale
Sensitive demographics last	Place age, gender, income, race/ethnicity at the end	Reduces social identity threat (Steele & Aronson, 1995; Danaher & Crandall, 2008)
Open-ended before Likert	If both probe the same construct, ask open-ended first	Prevents scale anchoring of qualitative responses
Topic blocks before context questions	Ask topic-specific attitudes before general context questions	Reduces question-order priming
Randomize within blocks	Randomize item order within blocks when order is unconstrained	Prevents primacy/recency effects and straight-lining
Introduce constructs before measuring them	Brief orienting text before unfamiliar construct blocks	Reduces measurement error from misunderstanding
Pre-register order decisions	Document randomization plan before data collection	Prevents researcher degrees of freedom

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Part 10 — Version Control Log

Version	Date	Author	Changes made
v0.1 — draft			Initial item generation
v0.2 — post peer review			
v0.3 — post cognitive interview			
v0.4 — post pilot test			
v1.0 — final (pre-registration)			

References

• Abbey, J. D., & Meloy, M. G. (2017). Attention by design. *Journal of Operations Management*, 53–56, 63–70.

• Dylong, P., Setzepfand, P., & Uebelmesser, S. (2024). Priming attitudes toward immigrants: Implications for migration research and survey design. European Journal of Political Economy, 85, 102554. https://doi.org/10.1016/j.ejpoleco.2024.102554

• Galesic, M., & Bosnjak, M. (2009). Effects of questionnaire length. *Public Opinion Quarterly*, 73(2), 349–360.

• Jeong, D., Aggarwal, S., Robinson, J., Kumar, N., Spearot, A., & Park, D. S. (2023). Exhaustive or exhausting? Evidence on respondent fatigue in long surveys. Journal of Development Economics, 161, 102992. https://doi.org/10.1016/j.jdeveco.2022.102992

• Miller, A. (2025). Accessibility, disability, and inclusive instrument design: A critical literature review interrogating the user experience of online surveys and interviews. *The International Journal of Information, Diversity, & Inclusion*, *9*(1/2). https://doi.org/10.33137/ijidi.v9i1/2.43997

• Danaher, K., & Crandall, C. S. (2008). Stereotype threat in applied settings re-examined. *Journal of Applied Social Psychology*, 38(6), 1639–1655.

• Steele, C. M., & Aronson, J. (1995). Stereotype threat and the intellectual test performance of African Americans. *Journal of Personality and Social Psychology*, 69(5), 797–811.

• Tourangeau, R., Rips, L. J., & Rasinski, K. (2000). *The Psychology of Survey Response*. Cambridge University Press.